SANDEEP GUPTA KOLETI

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PROFESSIONAL SUMMARY

- Software engineering technical leader with 14+ years building enterprise platforms, distributed services, and full-stack applications across Oracle, Morgan Stanley, SMBC, and ADP. Combines Python, Java, Go, React/TypeScript, AWS, and Azure with hands-on architecture for new applications, financial analytics, and AI-enabled engineering workflows.
- Led technical delivery for five associates, recruited to double team strength, and mentored engineers through architecture and implementation. Built agent platformsd adopted by three additional teams and AI-assisted workflows eliminating 5,000 annual manual hrs. Experienced in backlog prioritization, secure integrations, deployment, and production reliability.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Oracle | Austin, TX Aug 2024 - Present

- Led scalability engineering for Oracle Fusion authorization services across transactional control, low-latency decision, and durable reconciliation planes; supported growth from 4,000 to 25,000+ production service pods across global tenancies.
- Optimized and hardened the authorization decision hot path by preserving database-free request processing and improving bulk policy evaluation, request limits, atomic policy-snapshot replacement, strict cache validation, and freshness aware readiness; reduced P99 latency by 40% while containing stale-policy and partial-load risk
- Architected and built a multi-agent orchestration platform from the ground up, with dependency-aware planning, live capability discovery, and A2A/ACP/MCP/REST integrations; designed concurrent execution, retries, timeouts, human approvals, and tracing. [Python, PydanticAI, LangGraph, LangChain]
- Enabled adoption by three additional teams for developer productivity, on-call automation, and cross-platform operational mandates; built a TypeScript workflow graph and persistent run history for inspectable execution.
- Owned platform backlog prioritization, breaking complex initiatives into actionable architecture, implementation, and production-stability work to guide engineering delivery.
- Applied RAG and vector search to historical-incident retrieval for on-call diagnosis; evaluated retrieval quality and response groundedness. [Python, Oracle Database 26ai Vector Search, Ragas, LangSmith]
- Selected for the cross-organizational AI V-team; designed and developed a self-service authorization assistant grounded in current roles, policies, and access-denial data. Preserved authorization controls and supported rollout to 10,000+ Fusion clients, reducing escalations and calls by 30%. [Oracle Fusion AI Agent Studio, MCP]
- Architected an AI-assisted performance-testing platform with three agents for scenario generation, experiments, and reporting; integrated MCP metrics, logs, and source correlation with deterministic validation, load limits, and baseline comparisons. [Python, TypeScript, Chaos/Gatling, Prometheus]
- Contributed to 60+ human-in-the-loop Codex skills spanning incident triage, engineering delivery, CI/CD, and release validation, with approval gates and post-action verification; AI-assisted workflows eliminated approximately 5,000 annual manual hours and reduced promotion failures by 95%.
- Designed and developed FacadeSDK, AuthNSDK, and CodeGenSDK integrations and customized AuthNSDK for multi-tenant workloads, API-key validation, and mTLS. [Java, Helidon]
- Developed Java/Helidon APIs and JDBC/UCP persistence for a Go-to-Java management-service migration. Implemented JAX-RS resource, service, and repository layers, prepared SQL, JSON/CLOB mapping, and atomic deployment/jobs.
- Standardized Kubernetes/Helm delivery for three services across 10+ cloud tenancies and regions, including workload identity, least-privilege RBAC, health checks, autoscaling, and rollout procedures;replicated infrastructure with Terraform.
- Drove production root-cause analysis across API, database, and deployment failures; instrumented services with metrics, distributed traces, structured logs, alerts, and runbooks, reducing MTTR by 70%. [Prometheus, Mimir, OpenTelemetry]
- Audited approximately 150 SQL queries, analyzed execution plans, identified concurrency defects, and expanded connection-pool telemetry to guide query, index, and schema remediation. [SQL, Oracle Database]

Senior Software Engineer | SMBC | Jersey City, NJ Sep 2023 - April 2024

- Architected counterparty-risk analytics services and a full-stack portal; improved throughput by 60% through caching, compression, and rate limiting. Integrated analytics using Azure Databricks and Spark MLlib. [Python, Django, React, TypeScript, Azure, Kubernetes, Redis]
- Built data-ingestion microservices and stored procedures; integrated Denodo GraphQL APIs and Azure APIs using Entra ID authentication, with Azure DevOps delivery. [Azure Durable Functions, SQL, Oracle Database, SQL Server]

Technical Lead | Morgan Stanley | New York, NY Oct 2019 - Jul 2023

- Led technical delivery and prioritized backlogs for analytics and enterprise applications; recruited engineers to grow team strength by 100% and mentored the expanded team through architecture, implementation, and delivery.
- Created DataAI, a self-service search-based analytics platform for wealth-management financial advisors and bank associates; built custom reporting and secure workflows with authentication, authorization, and rate limiting. [Python, React, Step Functions, Tableau, Dataiku]
- Developed an analytics and reporting portal with ML/NLP applications for anomaly detection, sentiment analysis, and topic modeling, reducing support calls by 35% and costs by $5M; housed 10+ analytical forms and 25+ Tableau reports for leadership visibility. [Python, scikit-learn, NLTK, Flask, SQL]
- Led five associates in building Agent Assist for client-service answers, training, and cross-business support, reducing average call time by 70%; led Amelia chatbot implementation for employee ticket creation and resolution. [Google Dialogflow, Python, conversational AI]
- Built incident forecasting and categorization applications to identify issues proactively and prevent escalations; developed MSRB/Bloomberg ingestion pipelines for trading analytics. [Python, ML, NLP, Flask, Spark, Kafka, SQL]

Lead Development Engineer | ADP | Florham Park, NJ Jul 2016 - Oct 2019

- Built ADP DataCloud's Insights Engine, applying correlations, rankings, and comparisons to 600K clients' payroll/time data; delivered personalized insights and mobile notifications, increasing adoption by 40%. [Python, Spark, ML]
- Developed full-stack Revenue Analysis and Revenue Reconciliation applications; applied Python and ML to financial warehouse data and revenue reports to identify gains, declines, and underlying drivers and inform business decisions.
- Selected for the Emerging Leaders Program; developed Smartform, a predictive Q&A application, and Smart Agents to flag unsatisfactory client interactions for service managers, reducing negative-sentiment cases by 40%. [Python, Flask, NLP, React, Node.js]

Software Development Engineer 2 | Infinera | Bengaluru, India / Sunnyvale, CA Jan 2012 - Jun 2014

- Developed sales analytics within the pricing team, supporting 20% higher profitability; built enterprise integrations reducing supply-chain lead time by 70%. [Java, MySQL, Azure, BizTalk, EDI]

Software Engineer | Reflexis Systems | Chennai, India Oct 2010 - Dec 2011

- Re-engineered forecasting and workforce-scheduling capabilities; developed a custom sales model supporting a $5M GameStop launch. [Java, MySQL]

TECHNICAL SKILLS

AI and Automation: OpenAI APIs, LangGraph, LangChain, PydanticAI, MCP, RAG, vector search, prompt engineering, Ragas, LangSmith, human approvals, Codex, Claude

Engineering and Data: Python, Java, Go, FastAPI, Flask, Django, React, TypeScript, Node.js, REST APIs, Oracle Database, SQL Server, MySQL, Redis, Kafka, Spark MLlib, Azure Databricks, Tableau

Cloud and Delivery: AWS (Lambda, S3, API Gateway, Step Functions); Azure, OCI; Terraform, Docker, Kubernetes, Helm, Git, GitLab, Jenkins, Azure DevOps, CI/CD, Agile/SDLC, JUnit, Mockito

Architecture and Reliability: Distributed systems, microservices, multi-tenancy, RBAC, mTLS, OpenTelemetry, Prometheus, root-cause analysis, production support

PERSONAL ENGINEERING PROJECTS

- **CloudOps AI:** Built an AWS-native multi-agent system for incident classification, root-cause analysis, and remediation using Bedrock AgentCore Runtime, Gateway (MCP), Lambda, CloudWatch, and Bedrock Knowledge Bases.
- Integrated Step Functions, DynamoDB, and AgentCore Observability for approval workflows, state persistence, recovery, remediation verification, and distributed tracing.

EDUCATION

UNIVERSITY OF NORTH CAROLINA , Chapel Hill, NC May 2016
Master of Business Administration (MBA), Analytics

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE (BITS), Pilani, India Jul 2010
Bachelor's (Honors), Mechanical Engineering Merit Award from the State Governor

Selected certifications and Training: Oracle Agentic AI; OCI Generative AI; OCI AI Vector Search; Oracle Fusion AI Agent Studio; OCI Data Science; Machine Learning; Natural Language Processing, Azure AI Associate, Java and J2EE Developer